Special Unitary Parameterized Estimators of Rotation (SUPER)

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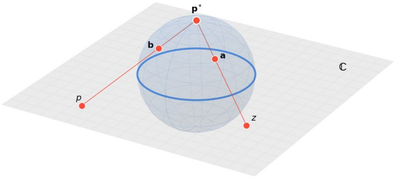
Special Unitary Group *SU(2)*



SU(2) consist of 2x2 unitary matrices with determinant 1 that describe rotations of the sphere in 3D. Closely related to quaternions, they act on 2D complex projective space via a stereographic projection of the sphere are an underutilized rotation representation in engineering.

$$\mathbf{U} = \begin{bmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{bmatrix}$$

$$\alpha\bar{\alpha} + \beta\bar{\beta} = 1, \quad \alpha, \beta \in \mathbb{C}$$

$$\psi_{\mathbb{C}}(\mathbf{a}): \ \frac{x_s}{1+z_s} + \frac{y_s}{1+z_s} i = x_p + y_p i = z$$

$$\mathbf{\Phi_U}: \ z \mapsto z' = \frac{\alpha z + \beta}{-\bar{\beta} z + \bar{\alpha}}$$

Wahba's Problem and Deep Learning

Wahba's problem seeks the optimal rotation aligning two sets of points on the sphere and is related to the problem of finding the nearest rotation matrix (in the Frobenius sense). Neural networks struggle to smoothly learn structured outputs such as rotations directly, so modern approaches predict higher-dimensional representations that are projected onto the rotation manifold and can be interpreted as solving Wahba's problem.

$$\min_{\mathbf{R} \in SO(3)} \sum_i w_i ||\mathbf{b}_i - \mathbf{R}\mathbf{a}_i||^2$$

Solutions to Wahba's Problem via *SU(2)*

Stereographic

$$w_i' = \frac{4w_i}{(1+x_i^2+y_i^2)(1+m_i^2+n_i^2)}$$

$$\sqrt{w_i'} \begin{bmatrix} x_i - m_i & -y_i - n_i & 1 + m_i x_i - n_i y_i & m_i y_i + n_i x_i \\ y_i - n_i & x_i + m_i & m_i y_i + n_i x_i & 1 - m_i x_i + n_i y_i \end{bmatrix} \mathbf{q}$$

3D

$$\sqrt{w_i} \begin{bmatrix} 0 & x_i - m_i & y_i - n_i & z_i - p_i \\ m_i - x_i & 0 & -z_i - p_i & y_i + n_i \\ n_i - y_i & z_i + p_i & 0 & -x_i - m_i \\ p_i - z_i & -y_i - n_i & x_i + m_i & 0 \end{bmatrix} \mathbf{q}$$

Möbius Transformation

$$\mathbf{m} = vec(\mathbf{M}) = [\sigma \quad \xi \quad \gamma \quad \delta]^T$$

$$[-z_i \quad -1 \quad p_i z_i \quad p_i]\,\mathbf{m}$$

Stereographic and 3D solutions estimate optimal quaternion **q**. Möbius transformation solution **m** is an approximation, requiring projection to *SU(2)*. Solutions obtained via SVD of stacked constraints above.

Linear constraints from the solutions provide useful forms for optimization, facilitating constrained optimization and implicit representations of aligning rotations. They enable the formulation of the efficient two-point Wahba's problem solutions below.

$$\mathbf{n}_1 = \mathbf{a}_1 \times \mathbf{a}_2, \quad \mathbf{n}_2 = \sqrt{\frac{||\mathbf{a}_1 \times \mathbf{a}_2||^2}{||\mathbf{b}_1 \times \mathbf{b}_2||^2}}(\mathbf{b}_1 \times \mathbf{b}_2), \quad \tilde{\mathbf{q}}_i = \begin{bmatrix} (\mathbf{a}_i + \mathbf{b}_i) \cdot (\mathbf{n}_1 - \mathbf{n}_2) \\ (\mathbf{a}_i - \mathbf{b}_i) \times (\mathbf{n}_1 - \mathbf{n}_2) \end{bmatrix}$$

$$\tau = (w_1 - w_2)||\tilde{\mathbf{q}}_1||^2||\tilde{\mathbf{q}}_2||^2, \quad \omega = 2w_1||\tilde{\mathbf{q}}_2||^2(\tilde{\mathbf{q}}_1 \cdot \tilde{\mathbf{q}}_2)$$

$$\nu = 2w_2||\tilde{\mathbf{q}}_1||^2(\tilde{\mathbf{q}}_1 \cdot \tilde{\mathbf{q}}_2), \quad \mu = \tau + \sqrt{\tau^2 + \omega\nu}$$

$$\mathbf{q} = \frac{\mu\tilde{\mathbf{q}}_1 + \nu\tilde{\mathbf{q}}_2}{\sqrt{||\tilde{\mathbf{q}}_1||^2\mu^2 + ||\tilde{\mathbf{q}}_2||^2\nu^2 + 2(\tilde{\mathbf{q}}_1 \cdot \tilde{\mathbf{q}}_2)\mu\nu}}$$

Weighted

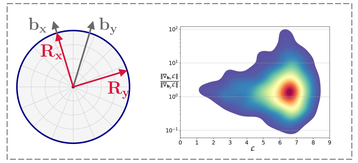
$$\mathbf{s}_1 = \mathbf{a}_1 + \mathbf{a}_2, \quad \mathbf{s}_2 = \sqrt{\frac{1 + \mathbf{a}_1 \cdot \mathbf{a}_2}{1 + \mathbf{b}_1 \cdot \mathbf{b}_2}}(\mathbf{b}_1 + \mathbf{b}_2)$$

$$\mathbf{d}_1 = \mathbf{a}_1 - \mathbf{a}_2, \quad \mathbf{d}_2 = \sqrt{\frac{1 - \mathbf{a}_1 \cdot \mathbf{a}_2}{1 - \mathbf{b}_1 \cdot \mathbf{b}_2}}(\mathbf{b}_1 - \mathbf{b}_2)$$

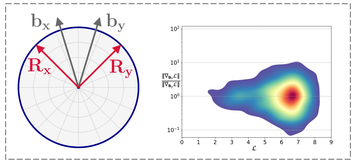
$$\tilde{\mathbf{q}} = \begin{bmatrix} (\mathbf{s}_1 + \mathbf{s}_2) \cdot (\mathbf{d}_1 - \mathbf{d}_2) \\ (\mathbf{s}_1 - \mathbf{s}_2) \times (\mathbf{d}_1 - \mathbf{d}_2) \end{bmatrix}$$

Unweighted

2-vec Representation (Efficient)



Gram-Schmidt



2-vec

$$\mathbf{b}_y' = \frac{||\mathbf{b}_x||}{||\mathbf{b}_y||}\mathbf{b}_y, \quad \mathbf{b}^+ = \frac{\mathbf{b}_x + \mathbf{b}_y'}{||\mathbf{b}_x + \mathbf{b}_y'||}, \quad \mathbf{b}^- = \frac{\mathbf{b}_x - \mathbf{b}_y'}{||\mathbf{b}_x - \mathbf{b}_y'||}$$

$$\mathbf{R} = \left[\tfrac{1}{\sqrt{2}}(\mathbf{b}^+ + \mathbf{b}^-), \ \tfrac{1}{\sqrt{2}}(\mathbf{b}^+ - \mathbf{b}^-), \ \mathbf{b}^- \times \mathbf{b}^+\right] \in SO(3)$$

Based on the unweighted two-point solution to Wahba's problem, 2-vec maps a 6D network prediction of coordinate axes to a rotation matrix. It provides a more balanced projection than commonly-used Gram–Schmidt, enabling better learning (outperforming GS on 41/52 metrics) for the same dimensionality and compute.

QuadMobius Representation (Performant)

$$\mathbf{G}_M(\Theta) = \begin{bmatrix} \theta_1 & \theta_2 + \theta_3 i & \theta_4 + \theta_5 i & \theta_6 + \theta_7 i \\ \theta_2 - \theta_3 i & \theta_8 & \theta_9 + \theta_{10} i & \theta_{11} + \theta_{12} i \\ \theta_4 - \theta_5 i & \theta_9 - \theta_{10} i & \theta_{13} & \theta_{14} + \theta_{15} i \\ \theta_6 - \theta_7 i & \theta_{11} - \theta_{12} i & \theta_{14} - \theta_{15} i & \theta_{16} \end{bmatrix}$$

$$\min_{\mathbf{m}} \mathbf{m}^H \mathbf{G}_M \mathbf{m} \ \ s.t. \ \ ||\mathbf{m}|| = 1$$

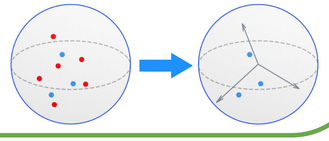
Algebraic

$$\mathbf{M}^* = \sqrt{\frac{\overline{det(\mathbf{M})}}{|det(\mathbf{M})|(2|det(\mathbf{M})| + Tr(\mathbf{M}^H\mathbf{M}))}}\mathbf{M}$$

$$\mathbf{Q} = \mathbf{M}^* + adj(\mathbf{M}^*)^H \in SU(2)$$

SVD

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

$$\mathbf{Q} = \sqrt{det(\mathbf{U}\mathbf{V}^H)}\mathbf{U}\mathbf{V}^H \in SU(2)$$

Based on the Möbius transformation approximate solution, QuadMobius maps a 16D network prediction to a Hermitian matrix, extracts the eigenvector with smallest eigenvalue, and projects it to *SU(2)*, achieving state-of-art performance on several benchmarks. Its two-stage procedure provides a robust and structured path for learning.